

Test & Measurement

What's Coming Up Next

While today's technology offers solutions for monitoring or remote reporting via mobile devices, test organisations need new expertise to unite the networking, Web services and mobile app portions of the solution



PXI platforms by National Instruments

UMA GUPTA

Testing plays a critical role from the very beginning of the product life-cycle, i.e., the design stage. Be it any technology or any product, unless and until it is validated and tested, it cannot be manufactured and produced for the market.

Test engineers need to measure the parameters of the signals with a high degree of accuracy and performance. Today, circuits are packed in compact assemblies and affected by a lot of interference, noise and unwanted signals. This, in turn, affects the measure-

ments. Instruments are being designed to measure such signal levels. The measurement data also needs to be analysed. To enable this, PC connectivity has become a common feature on general-purpose test instruments.

T&M needs across different segments are broadly common. But deep-down, there are different levels of testing and different-grade T&M products being used.

Requirements across segments

Within education, where test and measurement is mainly meant for

teaching or research purpose, there is quite a significant difference between two sub-segments. In undergraduate labs, basic or mid-range instruments such as digital multimeters, power supplies, oscilloscopes and function generators are mainly used to do some basic measurements and learn different aspects of testing. But in research work, high-end scopes, spectrum analysers, network analysers and vector signal generators are used in such areas as high-energy physics, nanotechnology, millimeter-wave and UWB.

In aerospace and defence, high-bandwidth, high-frequency T&M